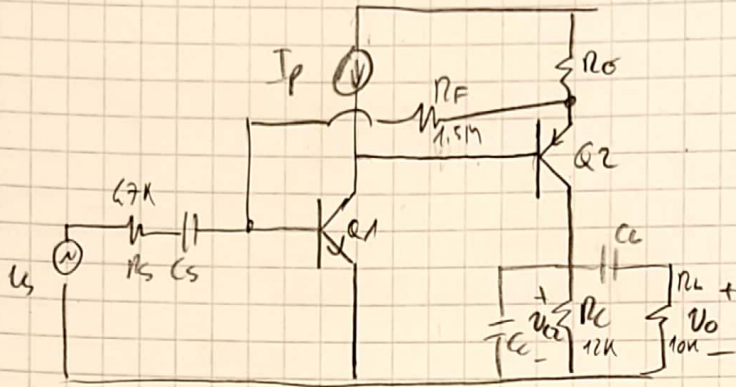


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9:25

 $V_{CC} = 26V$ 12) R_E for $V_{C2} = 11V$

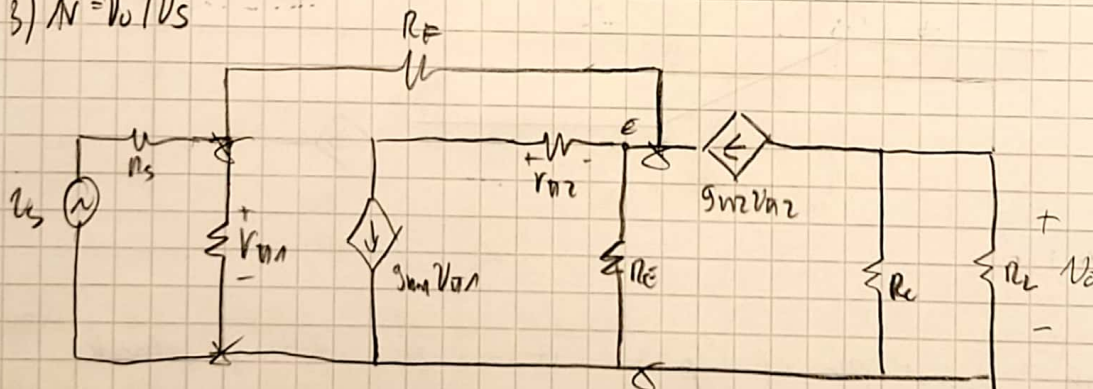
$$I_{C2} = \frac{V_{C2}}{R_C} = 0,917 \text{ mA} \quad I_{E2} = I_{C2} + \frac{I_{C2}}{\beta_{E2}} = 0,9258 \text{ mA}$$

$$I_{B1} = I_{P1} + I_{B2} = 1,009 \text{ mA} \quad V_{E2} = V_{BE1} + R_E I_{B1} = 15,83 \text{ V}$$

$$R_E = \frac{V_{CC} - V_{E2}}{I_{E2}} = 8,82 \text{ k}\Omega$$

$$V_{CE1} = V_{E2} + V_{BE2} = 15,13 \text{ V} \quad V_{CE2} = V_{C2} - V_{E2} = -4,83 \text{ V}$$

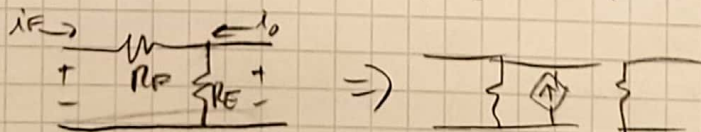
$$R_E = 8,82 \text{ k}\Omega \quad Q_1 (15,13 \text{ V}, 1,009 \text{ mA}) \quad Q_2 (-4,83 \text{ V}, 0,917 \text{ mA})$$

3) $A = V_o / V_s$ we have current mixing since $i_q = i_s + i_F$ open output port feedback is killed $\rightarrow$ current sampling. here β has to be positive and adimensional

$$i_F = \beta i_o$$

$$i_F = \beta i_o + V_F / R_F$$

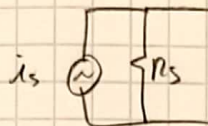
$$V_o = R_o i_o$$



$$\beta = \frac{i_F}{i_o} \Big|_{V_F=0} = - \frac{R_C}{R_F + R_E} \quad R_{if} = R_F + R_E = 1509,2 \text{ k}\Omega$$

$$R_{of} = R_C \parallel R_E \approx 8,15 \text{ k}\Omega$$

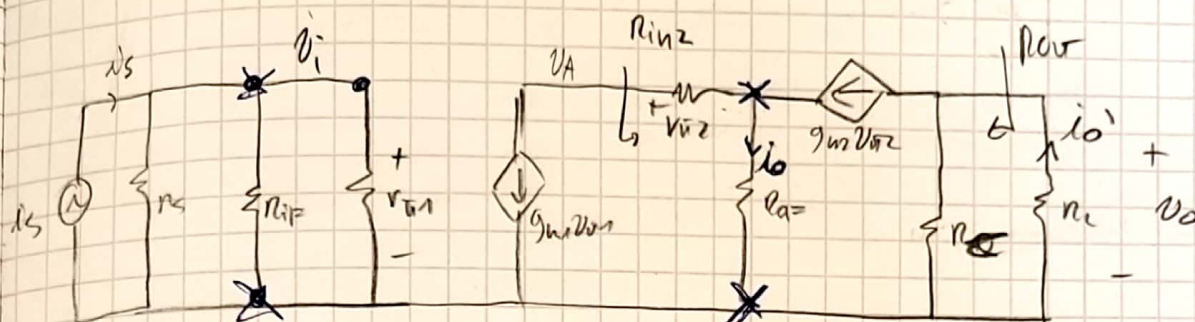
Norton an input



$$\frac{i_s}{v_s} = \frac{1}{R_s}$$

$$g_{m1} = 60,36 \mu A/V \cdot v_{i1} = 2,67 \mu A$$

$$g_{m2} = 36,68 \mu A/V \cdot v_{i2} = 2,72 \mu A$$



$$A_v = \frac{i_o}{i_s}$$

$$R_{i2} = 825,7 \mu A$$

sampled variable

is at the

β return output

port, voltage divider

$$i_o = -g_{m1} v_{i1} + g_{m2} v_{i2} = -g_{m1} v_{i1} + g_{m2} v_{i2} (-g_{m1} v_{i1})$$

$$i_s = \frac{v_{i1}}{R_s \parallel R_{i1} \parallel R_{i2}} = -g_{m1} v_{i1} - g_{m1} v_{i1} \beta_{o2}$$

$$\frac{i_o}{i_s} = \frac{v_{i1} (-g_{m1} - g_{m1} \beta_{o2})}{v_{i1}} (R_s \parallel R_{i1} \parallel R_{i2})$$

$$= -g_{m1} (1 + \beta_{o2}) (R_s \parallel R_{i1} \parallel R_{i2}) = -3660 \mu A$$

$$A_v^F = \frac{A_{v0}}{1 + \beta_{o2}} = \frac{-168,1 \mu A}{1 + 19,5} = -11,9 \mu A$$

$$A_v = \frac{v_o}{v_s} = \frac{i_s}{v_s} \cdot \frac{i_o}{i_s} \cdot \frac{v_o}{i_o} = \frac{1}{R_s} \cdot A_v^F \cdot (R_{o1} \parallel R_L) = -11,9 \mu A$$

$$1) R_{o1} = R_{o1} \parallel R_L = 5,65 \mu A$$

$$R_{o1}^F = R_{o1} \text{ feedback doesn't touch it}$$

$$R_{o1} = \frac{1}{\frac{1}{R_{o1}^F} - \frac{1}{R_L}} = 11,9 \mu A$$

$$2) H_F \text{ due to } C_c = S p_1^F$$

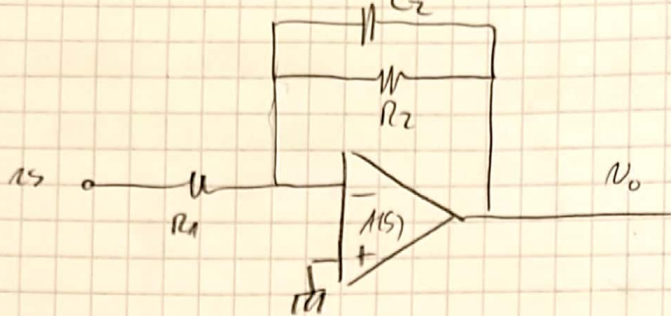
$$R_c^0 = R_{o1}^0 = 5,65 \mu A$$

$$f_L^0 = f_L^F = \frac{1}{2\pi R_c^0 C_c} = 5,86 \text{ MHz}$$

low frequency
caps are shorted on ground
from long time

feedback doesn't influence the resistance seen from
the cap

ESL)



1) expression of iden circuit gain

$$A_V(s) = -\frac{Z_2}{Z_1} \quad Z_2 = \frac{R_2}{1+sC_2R_2} \quad Z_1 = R_1$$

$$= -\frac{R_2}{1+sC_2R_2} \cdot \frac{1}{R_1} = -\frac{R_2}{R_1} \cdot \frac{1}{1+sC_2R_2}$$

2) \$e_m\$

$$T = A\beta = \frac{V_r}{V_e} = A(s) \cdot \frac{Z_1}{Z_2 + Z_1} = A(s) \cdot \frac{R_1}{\frac{R_2}{1+sC_2R_2} + R_1} = A(s) \cdot \frac{R_1(1+sC_2R_2)}{R_2 + R_1(1+sC_2R_2)}$$

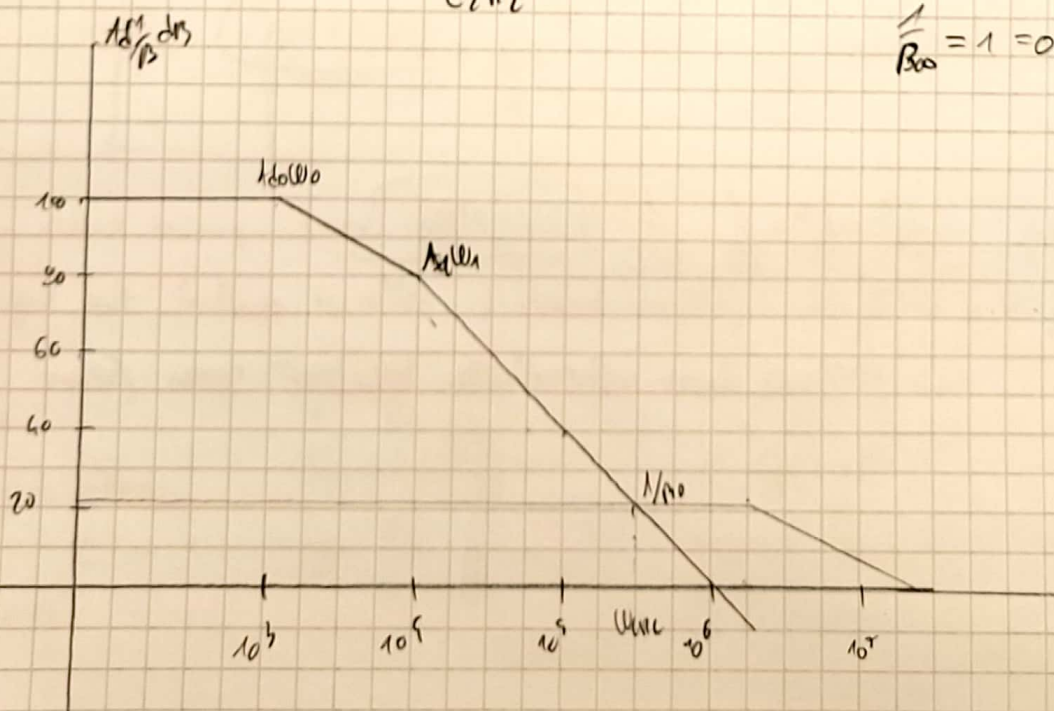
$$= A(s) \cdot \frac{R_1(1+sC_2R_2)}{R_2 + R_1(sC_2R_2R_1 + 1)} = \frac{A(s)R_1}{R_1 + R_2} \cdot \frac{1+sC_2R_2}{1+sC_2(R_1R_2)} \quad \text{dominant pole comp / (less 12g)}$$

$$\beta(s) = \frac{1+sC_2R_2}{1+sC_2(R_1R_2)} \quad \omega_p = \frac{1}{C_2(R_1R_2)} = 2,36 \cdot 10^7 \text{ rad/s}$$

$$\omega_z = \frac{1}{C_2R_2} = 2,13 \cdot 10^6 \text{ rad/s}$$

$$\frac{1}{\beta_0} = \frac{R_1R_2}{R_1} = 21 \text{ dB}$$

$$\frac{1}{\beta_{\infty}} = 1 = 0 \text{ dB}$$



C_2 has the wrong value, $t = \pm 40 \text{ dB} \rightarrow \varphi_m \rightarrow 0^\circ$

$$\left\{ \begin{array}{l} A_{d0} \omega_0 = A_1 \omega_1 \\ A_1 \omega_1^2 = 1/p_1 \omega_{crit}^2 \end{array} \right\} \quad \omega_{crit} = \sqrt{p_3 \omega_0 \omega_1 A_{d0}} = \boxed{3 \cdot 10^5 \text{ rad/s}}$$

$$\varphi_m = 180 - \arctan \frac{\omega_{cr}}{\omega_0} - \arctan \left(\frac{\omega_{cr}}{\omega_n} \right) - \arctan \left(\frac{\omega_{cr}}{\omega_p} \right) + \arctan \left(\frac{\omega_{cr}}{\omega_2} \right) = \boxed{9,4^\circ}$$

3) C_2 to have $\varphi_m = 45^\circ$

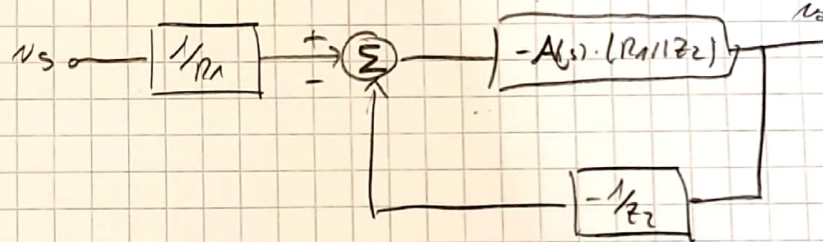
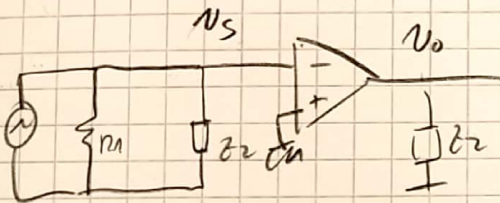
We need to have the zero of $1/\beta$ (pole of β) before the ~~zero~~ ω_{cr} in order to open and close at 20 dB

$$\omega_{cr} = \omega_{cr} \quad C_2 = \frac{1}{\omega_{cr} R_2} = \boxed{33,3 \text{ pF}}$$

4) BD feedback

inverting configuration $\rightarrow$ voltage sampling, current mixing

Norton equivalent, body effects $r_{1P} = R_1 = 1 \text{ k}\Omega$, $\beta = -1/R_2$



$$A_{OL} = \frac{v_o}{i_s} = \frac{v_o}{v_s} \cdot \frac{v_s}{i_s} = -A(s) (R_1 || R_2)$$

5) Input scaling function $K_{in} = \frac{1}{R_1}$

6) Voltage gain $A_V^F = \frac{v_o}{v_s} \text{ at } \omega = 0$

$$A_{OL}(0) = A(0) = -10^5 \cdot R_1 || R_2 = -909090,9 \text{ k}\Omega$$

$$A_V^F(0) = \frac{A_{OL}(0)}{1 + \beta(0) A_{OL}(0)} = \frac{-10^5 \cdot R_1 || R_2}{1 + 10^5 \cdot \frac{R_1 || R_2}{R_2}} \cdot \frac{1}{R_1} = \boxed{-9,9999}$$

$$A_V(0) = -\frac{R_2}{R_1} = -10$$

$$|\Delta A_V| = \left| \frac{A_V(0) - A_{Videal}}{A_{Videal}} \right| = \boxed{0,011\%} \quad \text{conv}$$